

Phase 18 Demo: Partial Function Operators

Example 1 : Partial injection operator

Basic function type with partial injection ($\succmapsto$) operator:

$$\left| \begin{array}{l} f : \mathbb{N} \succmapsto \mathbb{N} \\ \hline \text{dom } f = \{1, 2, 3\} \\ f(1) = 10 \\ f(2) = 20 \\ f(3) = 30 \end{array} \right|$$

Example 2 : Partial function operator

Basic function type with partial function ($\mapsto$) operator:

$$\left| \begin{array}{l} g : \mathbb{N} \mapsto \mathbb{N} \\ \hline \text{dom } g = \{1, 2\} \\ g(1) = 100 \\ g(2) = 200 \end{array} \right|$$

Example 3 : Complex types with partial injection

Using partial injection ($\succmapsto$) with custom types:

$Year == \mathbb{N}$

$[EntryAD]$

$TableAD == \text{seq } EntryAD$

$$\left| \begin{array}{l} recordsAD : Year \succmapsto TableAD \\ \hline \text{dom } recordsAD = \{1993, 1994, 1995\} \end{array} \right|$$

Example 4 : Mixed function type operators

Different function type operators:

$$\left| \begin{array}{l} total : \mathbb{N} \longrightarrow \mathbb{N} \\ partial : \mathbb{N} \mapsto \mathbb{N} \\ injection : \mathbb{N} \succmapsto \mathbb{N} \end{array} \right|$$